

CUSTOMER CHURN PREDICTION IN THE TELECOMMUNICATIONS INDUSTRY USING MACHINE LEARNING

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Data Science and AI

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DECLARATION

I am declaring that the work included in this project report is entirely my own. The analysis, the implementation, and the code were carried out by me during my participation in the IDRA Summer Training Program. All external sources, datasets, and references utilized in this project have been properly acknowledged.

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ABSTRACT

Customer churn is one of the key challenges faced by the telecommunications industry because losing customers can directly impact revenue and future business growth. This project aims to predict customer churn using the **Telco Customer Churn dataset**, which includes information about customers' demographics, services, contracts, tenure, and billing details.

The project follows a complete data science process, starting with data cleaning and preprocessing and continuing through exploratory data analysis, statistical analysis, feature engineering, and machine learning. The analysis revealed several factors associated with higher churn, particularly contract type, internet service, payment method, tenure, and monthly charges. Customers on month-to-month contracts, those using electronic checks, fibre optic service users, and customers with shorter tenures were found to have comparatively higher churn rates.

Five classification models were trained and evaluated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbour's (KNN), and Support Vector Machine (SVM). Based on the selected evaluation criteria, **SVM was chosen as the final model**, with an accuracy of **74.59%**, precision of **51.41%**, recall of **78.07%**, F1-score of **62.00%**, and ROC-AUC of **82.49%**.

Overall, the project shows how machine learning can be used to identify customers who are more likely to leave and can provide useful insights for developing targeted customer retention strategies.

Keywords: Customer Churn; Telecommunications; Machine Learning; Exploratory Data Analysis; Support Vector Machine

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1. INTRODUCTION

1.1 Background

Customer churn is a common issue in the telecommunications industry. When customers stop using a company's services, it can lead to a loss in revenue and also increase the cost of attracting new customers. For this reason, understanding why customers leave is important for telecom companies. By identifying the factors linked to churn, companies can recognize customers who may be more likely to leave and take steps to retain them.

This project focuses on **predicting customer churn** using the Telco Customer Churn dataset. The dataset provides information about customers, including their demographic details, subscribed services, contract types, tenure, and billing information. Machine learning methods are applied to these factors to determine whether a customer is likely to discontinue their service.

1.2 Context of the Problem

Telecommunications companies have a wide range of customers, and their needs, service choices, contract types, and payment methods can vary considerably. Some customers may decide to leave because of pricing, their service experience, contract conditions, or a combination of different factors.

The analysis carried out in this project can be useful for **telecom companies, customer-retention teams, and business decision-makers** by highlighting customer groups that show a greater tendency to churn. Using predictive models, businesses can identify these customers earlier and take preventive measures, such as providing personalized offers, better customer support, or more suitable service plans.

1.3 Scope of the Study

This study focuses on analysing and predicting customer churn using the Telco Customer Churn dataset. The work covers the major stages of the data science process, including **data cleaning, preprocessing, exploratory data analysis, statistical analysis, feature engineering, machine learning model development, model evaluation, and business recommendations**.

Five classification algorithms are considered in the study, and **Support Vector Machine (SVM)** is selected as the final model according to the evaluation criteria defined for the project.

The study is restricted to the information provided in the dataset. It does not consider real-time customer behaviour, external market conditions, customer satisfaction survey responses, or the actual implementation of the prediction model within a telecom company's operational system.

2. PROBLEM STATEMENT AND MOTIVATION

Problem Statement

How can machine learning be used to predict customer churn and determine which customer characteristics and service-related factors are linked to a higher likelihood of customers leaving a telecommunications service?

Motivation

Customer churn is a significant concern for telecommunications companies because losing existing customers can affect both revenue and long-term business growth. If customers who are likely to leave can be identified in advance, companies can take suitable steps to address their concerns and improve customer retention.

The main motivation behind this project is to develop a **data-driven approach for predicting customer churn**. The aim is to identify customers who may be at a higher risk of leaving by analyzing factors such as their demographic information, subscribed services, contract type, tenure, and billing details.

A reliable churn prediction model can help businesses:

- Identify customers who may be at risk of leaving at an early stage.
- Understand the major factors related to customer churn.
- Focus retention efforts on customers who need more attention.
- Make better decisions using customer data and predictive insights.
- Potentially reduce customer loss and improve overall customer retention.

3. OBJECTIVES

The main objectives of this project are:

1. **To understand and prepare the Telco Customer Churn dataset** by identifying data quality issues and preparing relevant customer, service, contract, and billing information for analysis.
2. **To identify important patterns and factors associated with customer churn** through exploratory and statistical analysis of customer characteristics and service-related variables.
3. **To develop machine learning classification models** to predict whether a customer is likely to churn.
4. **To evaluate and compare the predictive performance of different machine learning models** using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and confusion matrices.
5. **To select the most suitable churn prediction model** based on predefined evaluation criteria.
6. **To translate the analytical and predictive findings into actionable business recommendations** that can support customer retention and reduce potential churn.

4. RESEARCH QUESTIONS

This project is designed to answer the following research questions:

1. What are the main characteristics of the Telco Customer Churn dataset, and what data quality issues need to be addressed before analysis?
2. Which customer, service, contract, and billing-related variables show a significant relationship with customer churn?
3. How does the churn rate differ across important customer and service categories, including contract type, internet service, and payment method?
4. How do numerical variables, such as tenure and monthly charges, differ between customers who churn and those who remain with the service?
5. How well can different machine learning models predict whether a customer is likely to churn?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used for this project is the **Telco Customer Churn dataset**, which contains information about customers' demographics, subscribed services, contract details, tenure, and billing.

The dataset was provided as part of the project/internship resources and was used for academic and analytical purposes. The original customer records were not modified, except for the documented data-cleaning and preprocessing steps carried out during the analysis.

Source: Telco Customer Churn dataset (IDRA sample dataset)

Access Date: September 2026

Usage: The dataset was used strictly for educational, academic, and project analysis purposes.

5.2 Dataset Size and Structure

The dataset used in this project contains **7,043 customer records and 20 variables**. Each row represents an individual customer, while the columns contain information related to the customer's demographic characteristics, subscribed services, contract details, tenure, and billing information.

The variables were examined to understand their relevance to the analysis and their potential usefulness for predicting customer churn. The **Churn** variable was identified as the **target variable**, as it indicates whether a customer has discontinued the company's services. This is the outcome that the machine learning models are designed to predict.

Table 1 provides a summary of the key variables in the dataset and their roles in the analysis.

Column	Description	Role	Type
Customer ID	Unique identifier assigned to each customer.	Identifier	Categorical
Tenure	Number of months the customer has stayed with the company.	Predictor	Numerical
Contract	Type of contract chosen by the customer.	Predictor	Categorical
Monthly charges	Monthly amount charged to the customer.	Predictor	Numerical
Churn	Indicates whether the customer left the company.	Target variable	Categorical

Table 1. Summary of the Dataset

5.3 Data Dictionary

The following data dictionary presents the main variables selected for the analysis, including their meaning, data type, and role in the customer churn prediction task.

Variable name	Meaning	Data type	Role
Tenure	Number of months the customer has been with the company.	Integer	Feature

Monthly charges	Monthly amount charged to the customer.	Float	Feature
Total charges	Total amount charged to the customer.	Float	Feature
Contract	Type of customer contract.	Categorical	Feature
Churn	Indicates whether the customer discontinued the service.	Categorical	Target

Table 2. Data Dictionary

6. DATA CLEANING

Before starting the analysis, the dataset was checked for common data quality issues, including missing values, duplicate records, inconsistent categories, incorrect data types, and unusual values.

- **Missing Values:** After preprocessing, no missing values were found in the dataset, so no imputation was needed.
- **Duplicate Records:** The dataset was checked for duplicate entries, and no problematic duplicate customer records were retained.
- **Invalid or Inconsistent Values:** The categorical variables were reviewed to identify any inconsistent or invalid categories. Where required, the categories were standardized to maintain consistency.
- **Data Types:** The data types of the variables were examined and converted where necessary so that the data could be used correctly for analysis and machine learning.
- **Outliers:** Numerical variables were checked for unusually high or low values. No records were removed only because they were identified as outliers, since these values could represent actual customer behaviour and may provide useful information for predicting churn.

Overall, the cleaning process was carried out to maintain as much of the original information as possible while making sure that the dataset was suitable for exploratory analysis, statistical analysis, and machine learning.

6.1 Missing Values

A missing-value check was carried out for all the columns in the dataset to make sure that there were no incomplete entries.

- **Columns with missing values:** None
- **Total missing values:** 0
- **Percentage of missing values:** 0%
- **Treatment:** No imputation or removal of records was necessary.

Since the dataset contained no missing values after preprocessing, it was considered complete in this regard and was used for the next stages of the analysis.

6.2 Duplicate Records

The dataset was checked for duplicate records to ensure that the same customer information was not repeated.

- **Duplicate records found:** 0
- **Duplicate records removed:** 0
- **Change in dataset size:** None; the dataset remained at **7,043 records**.

Since no duplicate records were found, there was no need to remove any entries before proceeding with the analysis.

6.3 Data Types and Inconsistent Values

The dataset was checked to make sure that the variables had suitable data types and that the categorical values were consistent.

- **Data Type Correction:** The TotalCharges column was converted to a **numeric data type** so that it could be properly used for numerical calculations, analysis, and machine learning.
- **Categorical Values:** The categorical columns were reviewed for any incorrect, inconsistent, or invalid text values. No significant inconsistencies were found.
- **Result:** After these checks, the variables were in appropriate formats and the dataset was ready for further analysis and modelling

6.4 Outlier Handling

The main numerical variables were checked for possible outliers using the **Interquartile Range (IQR) method**.

- **Variables checked:** tenure, MonthlyCharges, and TotalCharges.
- **Method used:** The IQR method was applied, where values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are treated as potential outliers.
- **Detection:** Some extreme values were found in the numerical variables during the analysis.
- **Decision:** No observations were removed only because they were identified as outliers. These values may represent genuine customer characteristics, such as customers with long tenures or those paying higher monthly charges.
- **Justification:** Removing these observations could lead to the loss of useful information and might reduce the model's ability to recognize real customer churn patterns.

Therefore, the original observations were retained and used in the subsequent analysis and machine learning modelling.

6.5 Before vs After Cleaning

Table 3 summarizes the data quality status before and after the cleaning process.

Data quality check	Before cleaning	Action taken	After cleaning
Missing values	Present after TotalCharges conversion	Removed incomplete records	0
Duplicate records	0	No removal required	0
TotalCharges data type	Incorrect	Converted to numeric	Correct
Inconsistent categories	None significant	No correction required	Consistent
Outliers	Some potential outliers	Retained with justification	Retained

Table 3. Data Quality Before and After Cleaning

This shows that the cleaning process improved data consistency while preserving valid customer observations.

7. DATA PREPROCESSING

After cleaning the dataset, the data was prepared for machine learning by converting categorical variables into a numerical format, defining the target variable, scaling the numerical features, and dividing the data into training and testing sets.

- **Encoding:** Categorical variables were converted into numerical form using **One-Hot Encoding**. The first category was dropped to avoid creating redundant dummy variables.
- **Target Preparation:** The Churn column was converted into a binary target variable, where **0 represents No Churn** and **1 represents Churn**.
- **Feature Selection:** The customer identifier was not included in the modelling process because it does not provide useful predictive information. The remaining relevant customer, service, contract, tenure, and billing variables were used as features.
- **Feature Scaling:** The numerical variables tenure, MonthlyCharges, and TotalCharges were standardized using **StandardScaler** so that they were placed on a comparable scale.
- **Train-Test Split:** The dataset was divided into **80% training data** and **20% testing data**. Stratified sampling was used to maintain a similar proportion of churned and non-churned customers in both sets.

In the implemented workflow, **encoding and scaling were carried out before the train-test split**. The processed data was then divided into training and testing sets. The final training set contained **5,634 records**, while the testing set contained **1,409 records**.

7.1 Categorical Encoding

The dataset includes several categorical variables related to customer demographics, subscribed services, contract types, and payment methods. Since these variables are stored as categories rather than numerical values, they need to be converted into a format that machine learning models can process.

One-Hot Encoding was used to transform the categorical variables into numerical features. The `drop_first=True` option was applied so that one category from each variable was removed, helping to avoid redundant dummy variables and reduce multicollinearity.

One-Hot Encoding was appropriate for this dataset because the categorical variables are **nominal**, meaning their categories do not follow any natural order. This approach allows the models to use the categorical information without incorrectly assigning numerical rankings to different categories.

7.2 Feature Scaling and Transformation

The numerical variables tenure, MonthlyCharges, and TotalCharges were standardized using **StandardScaler**.

Scaling was necessary because these variables have different numerical ranges. If the features are left on their original scales, variables with larger numerical values can have more influence on certain machine learning algorithms.

Standardization changes the numerical features to a common scale, with a mean of approximately **0** and a standard deviation of approximately **1**. This is especially important for the **SVM model**, since SVM can be affected by differences in feature scales. Scaling is also useful for other models that depend on distances or optimization, such as **KNN and Logistic Regression**.

The categorical dummy variables were kept in their original **binary (0/1) form**.

7.3 Train-Test Split

The prepared dataset was divided into **80% training data and 20% testing data** using `random_state=42`. Stratification was also applied so that the proportion of churned and non-churned customers remained similar in both datasets.

- **Training set:** 5,634 records
- **Testing set:** 1,409 records
- **Split ratio:** 80:20
- **Random state:** 42

The train-test split allows the machine learning models to learn from one part of the dataset and then be evaluated on data that was not used during training. This provides a more realistic measure of how well the models can perform on unseen customer data.

Data Leakage

In the implemented workflow, **encoding and scaling were performed before the train-test split**. As a result, the `StandardScaler` was fitted using the complete dataset, including the test data. Strictly speaking, this introduces **minor data leakage**, because information from the test set was used to calculate the scaling parameters.

This is a methodological limitation of the current implementation. In an improved version of the workflow, the scaler should be **fitted only on the training data** and then used to transform the test data. This would prevent information from the test set from influencing the preprocessing step and provide a more reliable evaluation of model performance.

7.4 Final Feature Set

After preprocessing and encoding, the dataset contained **30 input features** that were used for machine learning.

Input Features

The final set of input features consisted of:

- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges
- One-hot encoded variables representing **gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, and PaymentMethod**.

Target Variable

The target variable used for prediction was:

- **Target:** Churn_Yes
- **0:** No Churn
- **1:** Churn

The customer identifier was excluded from the feature set because it is simply an identification field and does not contain meaningful information that would help predict customer churn.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was carried out to get a better understanding of the customer data and to identify patterns that may be related to customer churn.

The analysis mainly focused on:

- **Churn distribution:** Examining the proportion of customers who churned compared with those who continued using the service.
- **Numerical variables:** Studying variables such as tenure, MonthlyCharges, and TotalCharges to understand their distributions and general patterns.
- **Categorical variables:** Exploring customer and service-related categories to see how different groups are represented in the dataset.
- **Bivariate relationships:** Comparing customer and service characteristics between churned and non-churned customers to identify differences between the two groups.
- **Correlation analysis:** Examining relationships between numerical variables to understand how they are related to one another.

Only visualizations that provided useful information about **customer behaviour, churn patterns, and relationships between variables** were included in the analysis. The observations obtained from EDA were then used to support the later stages of **statistical analysis, feature engineering, and machine learning**.

8.1 Univariate Analysis

For the univariate analysis, the distributions of the three main numerical variables—**Tenure, MonthlyCharges, and TotalCharges**—were examined using histograms. These variables were selected because they directly describe customer duration and billing behavior, which are relevant to churn analysis.

```
plt.figure(figsize=(5,4))
plt.hist(df_analysis['tenure'], bins=20, edgecolor='black')
plt.title('Distribution of Tenure')
plt.xlabel('Tenure (Months)')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```

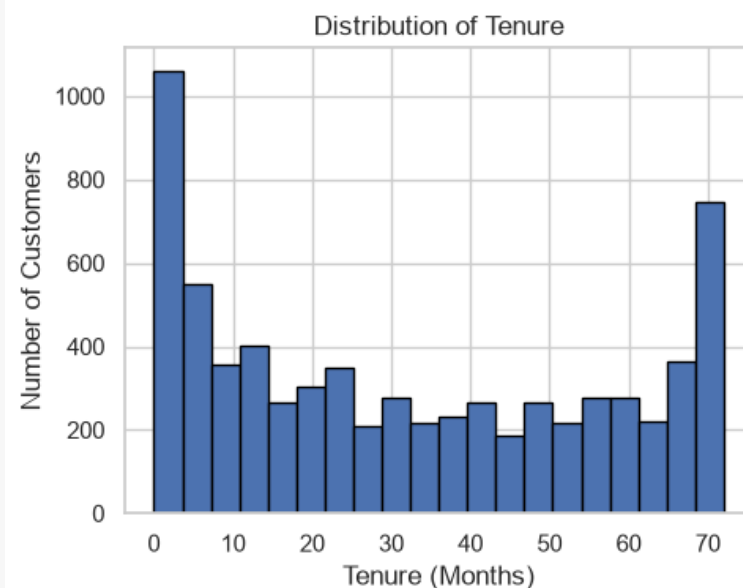


Figure 1. Distribution of Tenure

The tenure distribution is concentrated at lower tenure values, with another noticeable concentration among customers with longer tenure, particularly around **60–72 months**. This indicates that both newer and long-term customers are represented in the dataset. The distribution is not perfectly symmetric and shows variation across the customer lifecycle.

8.2 Categorical Analysis

Categorical analysis was carried out to understand how customers are distributed across different categories and whether churn rates vary between these groups. The analysis mainly focused on **contract type**, as it is an important factor that can be associated with customer churn.

A **count plot** was used to compare the number of churned and non-churned customers across the different contract categories. This comparison helps highlight differences in churn behaviour between customers with different types of contracts.

```
plt.figure(figsize=(4, 3))
sns.countplot(data=df_analysis, x='Contract', hue='Churn')
plt.title('Customer Churn by Contract Type')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```

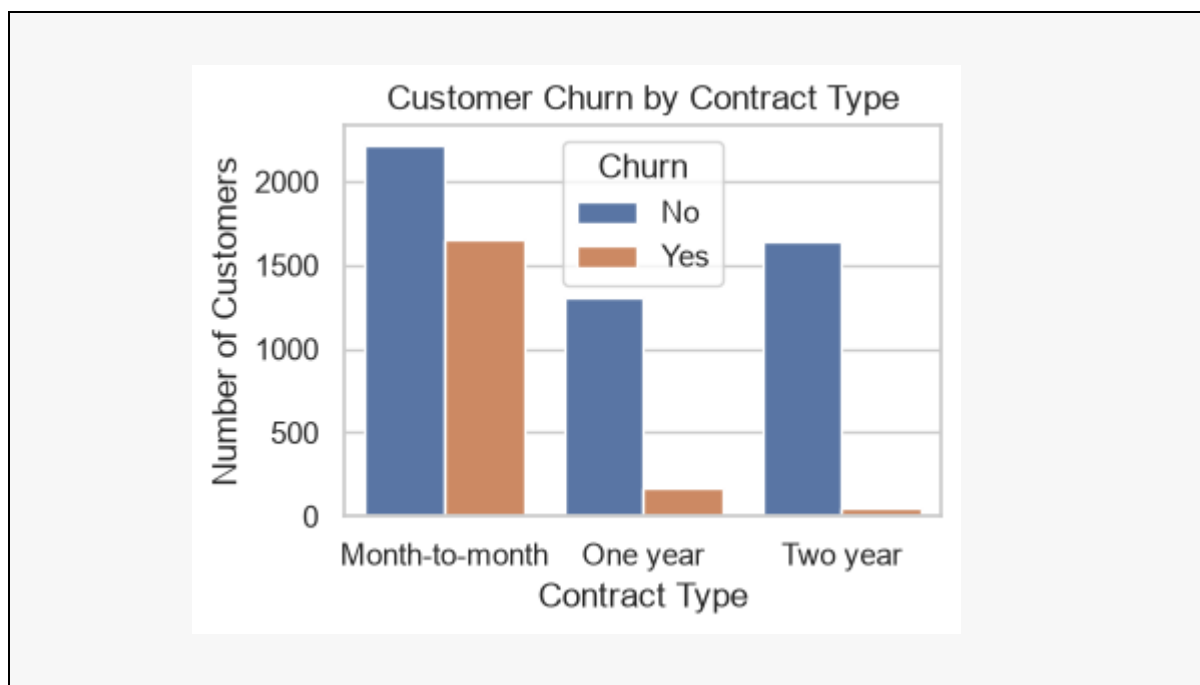


Figure 2. Customer Churn by Contract Type

The chart shows a clear difference in churn across contract types. Month-to-month customers have the highest number of churned customers, while one-year and two-year contract customers have substantially fewer churn cases. This indicates that contract type is an important categorical variable associated with customer churn.

8.3 Bivariate / Relationship Analysis

Bivariate analysis was carried out to understand the relationship between numerical variables and customer churn. Box plots were used to compare the distributions of numerical variables for customers who churned and those who stayed. The analysis mainly focused on **tenure**, as there is a noticeable difference in tenure between the two churn groups, providing useful insight into customer retention.

```
plt.figure(figsize=(4, 4))
sns.boxplot(data=df_analysis, x='Churn', y='tenure')
plt.title('Tenure Distribution by Churn Status')
plt.xlabel('Churn')
plt.ylabel('Tenure (Months)')
plt.tight_layout()
plt.show()
```

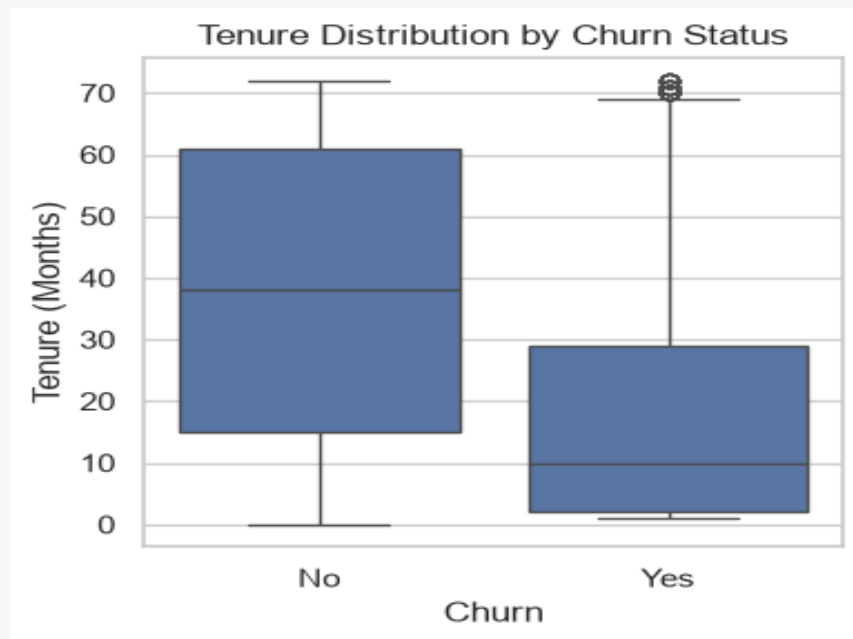


Figure 3. Relationship Between Tenure and Churn

The box plot shows that customers who churn generally have lower tenure than customers who do not churn. This indicates a negative relationship between customer tenure and churn tendency, with newer customers appearing more likely to churn. The difference is practically important because early-stage customers may require greater attention through onboarding and retention strategies. This relationship represents an observed association and does not by itself establish causation.

8.4 Correlation Analysis

Correlation analysis was carried out to understand the strength and direction of the relationships between the main numerical variables in the dataset. A correlation heatmap was used to present these relationships visually and to identify variables that could be useful for the machine learning models.

```
plt.figure(figsize=(7, 5))

corr = df_analysis[['tenure', 'MonthlyCharges', 'TotalCharges', 'SeniorCitizen']].corr()

sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')

plt.title('Correlation Heatmap of Numerical Variables')

plt.tight_layout()
plt.show()
```

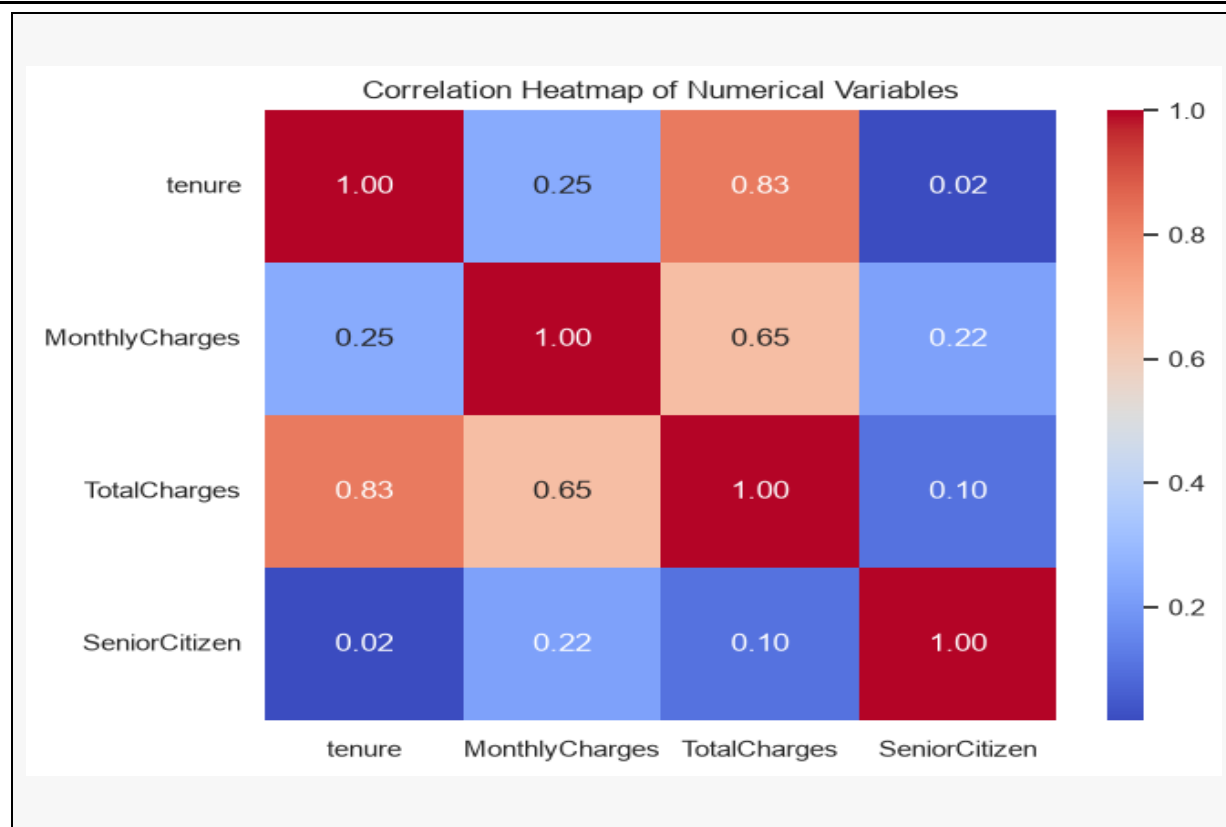


Figure 4. Correlation Heatmap of Numerical Variables

The heatmap shows a **strong positive correlation of 0.83** between **tenure** and **TotalCharges**, which means that customers who stay longer generally have higher total charges. **MonthlyCharges** and **TotalCharges** have a **moderate positive correlation of 0.65**, while the correlation between **tenure** and **MonthlyCharges** is **weaker at 0.25**. These relationships help in understanding how the numerical variables are connected and can also be useful when preparing the data for modelling. The strong relationship between tenure and TotalCharges indicates some level of feature redundancy, which should be kept in mind while interpreting the model results.

9. STATISTICAL ANALYSIS

Statistical analysis was carried out to summarize the main numerical variables and check whether important customer characteristics differed between customers who churned and those who stayed. Descriptive statistics and hypothesis tests were used to support the observations made during exploratory data analysis. The analysis included measures such as the **mean and standard deviation**, along with **independent-samples t-tests** and **chi-square tests** to examine differences and relationships between relevant variables.

Variable	Mean	Standard deviation	Interpretation
Tenure	32.37	24.56	Customers had an average tenure of about 32 months , with noticeable variation in how long they stayed.
MonthlyCharges	64.76	30.09	Monthly charges varied considerably across customers, reflecting differences in subscribed services.

TotalCharges	2,288.30	2,266.77	Total charges showed substantial variation due to differences in tenure and monthly billing
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Table 4. Descriptive Statistics of Key Variables

Table 4 presents the descriptive statistics for the main numerical variables. The results show noticeable differences in tenure, MonthlyCharges, and TotalCharges across customers. The variation in TotalCharges is expected, since it depends on both how long a customer has stayed with the company and the amount they are charged each month.

Statistical Test Results:

The statistical tests showed significant associations between several categorical variables and churn. **Contract type, internet service, and payment method** were significantly associated with churn, with all three chi-square tests producing **p-values below 0.05**. The independent-samples t-tests also showed significant differences in **tenure** and **MonthlyCharges** between churned and non-churned customers. Churned customers had a lower average tenure of **17.98 months**, compared with **37.57 months** for non-churned customers. In contrast, their average **MonthlyCharges** were higher, at **74.44** compared with **61.27** for customers who did not churn.

Overall, these findings provide statistical evidence of associations and differences that are relevant to churn prediction. However, statistical significance only indicates that an association or difference was observed; it does **not establish a causal relationship** between these variables and customer churn.

10. FEATURE ENGINEERING

Feature engineering was performed to transform the cleaned dataset into a suitable format for machine learning. Categorical variables were converted into numerical features using one-hot encoding, the target variable was prepared as a binary variable, and numerical features were standardized. These transformations ensured that the dataset was compatible with the selected classification algorithms.

Feature /Transformation	Method	Purpose	Role in model
Categorical variables	One-Hot Encoding	Converts categorical information into numerical features	Input features
Churn	Binary Encoding	Converts churn outcome into 0 = No Churn and 1 = Churn	Target variable
Tenure	StandardScaler	Places the numerical feature on a comparable scale	Input feature
MonthlyCharges	StandardScaler	Place the numerical feature on a comparable scale	Input feature
TotalCharges	StandardScaler	Places the numerical feature on a comparable scale	Input feature

Table 5. Engineered Features

Table 5 summarizes the feature transformations used before model development. One-hot encoding was applied to categorical variables, while **tenure**, **MonthlyCharges**, and **TotalCharges** were standardized using StandardScaler. The **Churn** variable was converted into a binary target for classification. These steps prepared the dataset for machine learning model development.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The problem is formulated as a **binary classification task**, where the goal is to predict whether a customer is likely to churn based on their demographic, service, contract, tenure, and billing characteristics.

The model uses **30 preprocessed input features** obtained after one-hot encoding and feature scaling. The target variable is **Churn_Yes**, where **0** represents **No Churn** and **1** represents **Churn**.

Classification is suitable for this problem because the target has only two possible outcomes. The models can learn patterns from existing customer information and use them to classify customers based on their likelihood of churning.

11.2 Model Selection

Five classification models were considered for the customer churn prediction task: **Logistic Regression**, **Decision Tree**, **Random Forest**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machine (SVM)**. These models were selected to compare different classification methods and determine which one performed best for churn prediction.

SVM was selected as the final model because it achieved the highest **F1-score of 62.00%**, which was the main evaluation criterion for this project. It also achieved a **recall of 78.07%**, meaning it was able to identify a large proportion of customers who were likely to churn.

Logistic Regression was another strong-performing model, with a slightly higher **recall of 78.34%** and **ROC-AUC of 84.18%**. However, its **F1-score of 61.43%** was lower than SVM's. Since F1-score was predefined as the primary model-selection criterion, SVM was selected despite Logistic Regression achieving slightly higher Recall and ROC-AUC.

11.3 Training the Model

```
from sklearn.svm import SVC
# Initialize the SVM model
svm_model = SVC(
    kernel='rbf',
    class_weight='balanced',
    random_state=42
)
# Train the model
svm_model.fit(X_train, y_train)
```

The **Support Vector Machine (SVM)** model was trained using the prepared training dataset. The **RBF (Radial Basis Function) kernel** was used to capture possible non-linear relationships between the input features and customer churn. The `class_weight='balanced'` parameter was used to give more weight to the churn class and help handle the class imbalance in the dataset. The model was trained using **X_train** as the input features and **y_train** as the target variable.

12. MODEL EVALUATION

The trained machine learning models were evaluated using the **unseen test dataset** to check how well they could predict customer churn. Since this is a binary classification problem, **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix** were used to evaluate model performance.

- **Accuracy:** Shows the overall percentage of customers correctly classified by the model.
- **Precision:** Shows how many of the customers predicted as churners actually churned.
- **Recall:** Shows how many of the actual churners were correctly identified.
- **F1-Score:** Combines Precision and Recall into a single measure and was used as the **primary criterion for selecting the final model**.
- **ROC-AUC:** Measures how well the model distinguishes between churned and non-churned customers across different classification thresholds.
- **Confusion Matrix:** Shows the counts of **true negatives, false positives, false negatives, and true positives**, giving a detailed view of the model's classification results.

12.1 Classification Evaluation

The trained classification models were evaluated on the **unseen test dataset** to assess how well they could predict customer churn. Since the project involves a binary classification problem, **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix** were used as the main evaluation measures.

Accuracy measures the proportion of customers that were classified correctly. **Precision** shows how many of the customers predicted as churners actually churned, while **Recall** shows how many of the actual churners were correctly identified. **F1-Score** provides a balance between Precision and Recall and was used as the **primary criterion for selecting the final model**. The **Confusion Matrix** gives a detailed view of the model's correct and incorrect classifications.

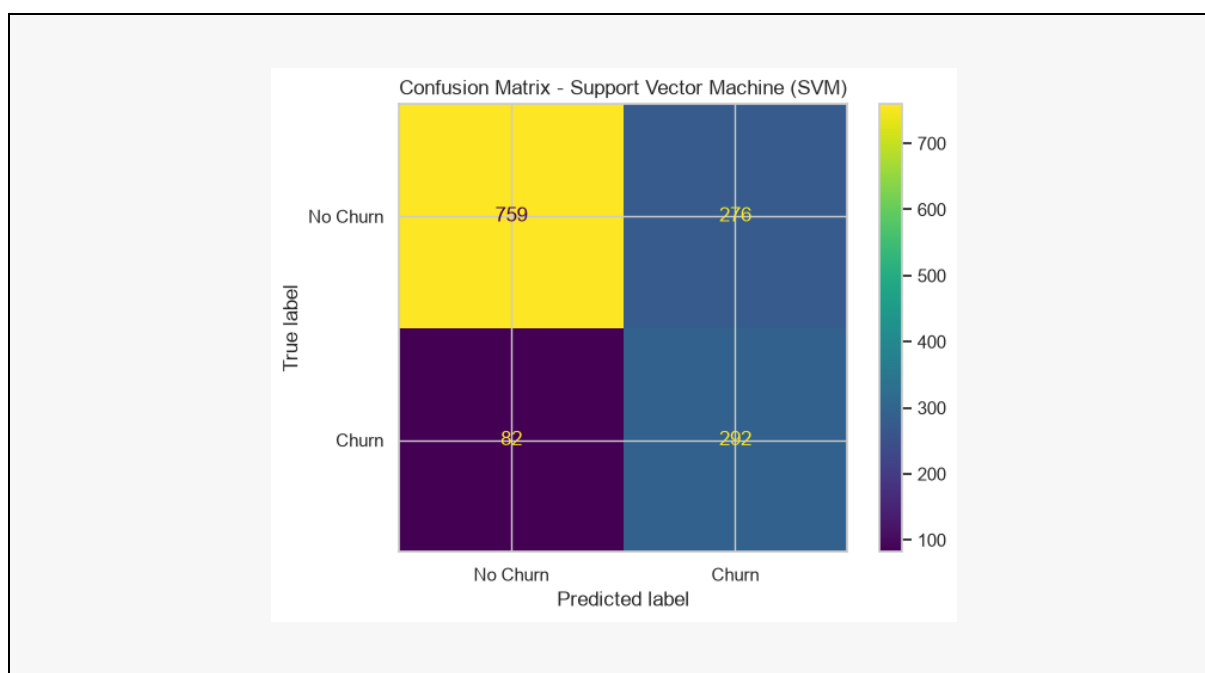


Figure 5. Confusion Matrix of the Final SVM Model

For the final SVM model, the confusion matrix contained the following results:

	Predicted No Churn	Predicted Churn
Actual No Churn	759	276

Actual Churn	82	292
---------------------	----	-----

The confusion matrix shows that the model correctly classified **759 non-churned customers** and **292 churned customers**. It also misclassified **276 non-churned customers as churners** and **82 churned customers as non-churners**. This shows that the model was able to identify most of the actual churners, although some incorrect predictions were still made.

The performance of all five classification models is presented in Table 7. The models were compared using Accuracy, Precision, Recall, and F1-Score.

Model	Accuracy	Precision	Recall	F1-score
Logistic regression	73.88%	50.52%	78.34%	61.43%
Decision tree	73.74%	50.55%	48.93%	49.73%
Random forest	77.00%	55.98%	62.57%	59.09%
K-Nearest neighbours (KNN)	76.30%	55.38%	55.08%	55.23%
Support Vector Machine (SVM)	74.59%	51.41%	78.07%	62.00%

Table 6. Classification Model Performance

Interpretation:

The results show that the five classification models performed differently across the evaluation metrics. **Random Forest** achieved the highest accuracy of **77.00%**, while **Logistic Regression** had the highest recall of **78.34%**. However, **SVM** achieved the highest **F1-Score of 62.00%**, giving it the best balance between Precision and Recall among the models tested.

The final SVM model achieved an **accuracy of 74.59%**, **precision of 51.41%**, **recall of 78.07%**, and **F1-Score of 62.00%**. Its high recall shows that the model was able to identify most customers who actually churned. However, the lower precision means that some customers predicted as churners did not actually leave.

Since **F1-Score was set as the primary criterion for model selection**, SVM was chosen as the final model for the customer churn prediction task.

12.2 Overfitting and Underfitting

The training and test performance of the final **Support Vector Machine (SVM)** model was compared to check how well the model fitted the data and whether there were signs of overfitting. The comparison of the training and test results is shown below.

Metric	Training performance	Test performance
Accuracy	76.78%	74.59%
Precision	54.15%	51.41%
Recall	81.61%	78.07%
F1-Score	65.10%	62.00%

The SVM model achieved an accuracy of **76.78%** on the training data and **74.59%** on the test data, giving a small difference of **2.19 percentage points**. Similarly, the F1-score was **65.10%** for the training data and **62.00%** for the test data, with a difference of **3.10 percentage points**.

The training and test results are fairly close, which suggests that the model is able to perform reasonably well on unseen data. Since there is no large gap between the two results, the model does not show strong

signs of overfitting. The slightly lower test performance is normal because the test data was not used during training.

Overall, the SVM model appears to be reasonably well-fitted and shows a small generalization gap. Further **cross-validation and hyperparameter tuning** could be used to verify and potentially improve its performance.

13. RESULTS AND FINDINGS

The analysis provided several important insights into customer churn in the telecommunications dataset. The main findings from the exploratory analysis, statistical tests, and machine learning models are summarized below.

1. Contract type is strongly associated with customer churn. Month-to-month customers had a much higher churn rate than customers with longer-term contracts. The churn rates were **42.71%** for month-to-month contracts, **11.27%** for one-year contracts, and **2.83%** for two-year contracts. The chi-square test also showed a statistically significant association between contract type and churn ($\chi^2 = 1184.60$, $p < 0.05$). This shows that contract type can be useful for identifying customers who may be at higher risk of leaving, particularly those on month-to-month contracts.

2. Customers with shorter tenure were more likely to churn. The analysis showed a clear difference in churn between newer and longer-term customers. Customers with **0–12 months of tenure** had a churn rate of **47.44%**, compared with **9.51%** for customers with **49–72 months** of tenure. The t-test also found a significant difference in tenure between churned and non-churned customers. The average tenure was **17.98 months** for churned customers and **37.57 months** for non-churned customers ($p < 0.05$). This suggests that the early stage of the customer relationship is an important period for retention efforts.

3. Internet service and payment method were associated with higher churn. Customers using **fiber optic service** had a churn rate of **41.89%**, while customers using **electronic check** as their payment method had a churn rate of **45.29%**. The chi-square tests showed significant associations between churn and both internet service ($\chi^2 = 732.31$, $p < 0.05$) and payment method ($\chi^2 = 648.14$, $p < 0.05$). These variables can therefore help in identifying customer groups with higher churn levels.

4. Churned customers generally had higher monthly charges. The average **MonthlyCharges** for churned customers was **74.44**, compared with **61.27** for customers who did not churn. The t-test showed that this difference was statistically significant ($t = -18.41$, $p < 0.05$). The EDA also showed that churned customers generally had higher monthly-charge values. This suggests that pricing and perceived value may be relevant to retention, although the results do not prove that higher charges directly cause customers to churn.

5. SVM provided the best balance between precision and recall. Among the five classification models, **SVM achieved the highest F1-score of 62.00%**, with an accuracy of **74.59%**, precision of **51.41%**, recall of **78.07%**, and ROC-AUC of **82.49%** on the test dataset. Random Forest achieved a higher accuracy of **77.00%**, while Logistic Regression had slightly higher recall (**78.34%**) and ROC-AUC (**84.18%**). However, SVM was selected as the final model because **F1-score was the predefined primary criterion**. Its relatively high recall means that it was able to identify a large proportion of customers who actually churned.

Overall Finding

Overall, the analysis shows that **contract type, tenure, internet service, payment method, and monthly charges** are important factors associated with customer churn. The final SVM model showed a reasonable ability to identify potential churners and can be used as a data-driven support for prioritizing customer-retention efforts. However, these results show **associations rather than causal relationships**, so the identified factors should not be interpreted as directly causing customer churn.

14. DISCUSSION

The results of this study can help understand what factors can influence customers' churn and how successfully machine learning can predict it. The analysis revealed that customers' contract type, tenure, internet service, payment method, and monthly charges are key predictors of churn.

A notably high customer churn rate was associated with month-to-month contracts and shorter tenure. It implies that the company should consider the need for taking measures to prevent such customers from leaving. For instance, it can try to improve customer acquisition by providing more support to new customers and those with no commitment to the company.

Another surprising result of the analysis was that fiber optic and electronic check are also associated with a higher probability of churn. It means that customers who choose these features are more likely to leave the company compared to others. However, the analysis only shows the correlation between the churn rates and these predictors. It does not prove the causal relationship between them and does not explain the potential mechanism of this relationship. Further studies with customers' feedback, service quality data, and other features may help identify the issue.

The results showed that churned customers had a higher mean of monthly charges and a lower mean of tenure. They can help assess the importance of price and commitment to the likelihood of churn. However, it does not mean that the company should solely focus on these aspects.

As for the model's results, they showed that the SVM algorithm had a relatively high F1-score (62%) and a recall (78.07%). Thus, the model can be a helpful tool for prioritizing customers at risk of leaving. Its precision was 51.41%, meaning that approximately half of the customers predicted as churners actually churned, while the remaining predictions were false positives. Therefore, it is important to keep in mind that the model only helps rank these customers and does not confirm their likelihood of churn.

In conclusion, the results can help telecommunication companies with reducing the churn rate. The analysis demonstrated that it can be done by identifying customers at risk of leaving and trying to provide them with more support. However, the model should only be used as an auxiliary tool to prioritize customers based on their likelihood of churn.

15. LIMITATIONS

The study has several limitations that should be noted when analyzing the results and applying them to other data sets or telecommunication companies' practice.

Dataset limitations refer to the fact that only variables that were present in the Telco Customer Churn dataset were used for analysis. Such aspects as the level of customer satisfaction, the presence of complaints, service quality, competitor offerings, and interaction history, among others, were not considered due to their unavailability in the chosen dataset.

Modelling limitations are explained by the choice of a simple model for the task. More complex algorithms could potentially yield better results, but they would require a more significant amount of time and data for training. The analysis used a single 80:20 train-test split, which might have resulted in overfitting or underfitting if multiple splits or cross-validation methods were used instead. In addition, hyperparameter optimization was not considered due to the same time-costliness and data redundancy.

The data preprocessing limitation is manifested in the fact that, when using the scale method, scaling did not occur within the train-test split. It could impact the final results as fitting the scaler on a single array, as opposed to the separate training and testing datasets, might introduce data leakage bias.

Finally, the generalization limitation is related to the Telco Customer Churn dataset being specific to a particular business or area. Thus, it is unclear whether the results and conclusions drawn from it would be

applicable to other companies or locations. The same limitation applies to deploying the final SVM model, which demonstrated strong recall but only moderate precision. In other words, some customers were identified as “likely churners” when, in fact, they were not. Therefore, the model should be used as an auxiliary instrument for identifying potential churners rather than a definitive assessment tool.

The study has several limitations; however, many of them are regarded as potential improvements for further analysis. Specifically, the next research could explore more options for improving customer retention. It could consider additional external factors, including the ones that were not available in the chosen dataset. The next study’s data collection phase could use more comprehensive tools, such as customer surveys, to better identify the target audience’s needs and pain points. Moreover, the data science part of the next research could use more data and time to optimize the hyperparameters and choose the best-performing algorithm for the given task.

16. CONCLUSION

This project aimed to predict customer churn in the telecommunications industry using a machine learning model and identify customer attributes associated with a higher likelihood of churn. The study implemented an entire data science workflow, ranging from data cleaning and preprocessing, descriptive and statistical analysis, feature engineering, model selection and training, and final evaluation.

The analysis revealed that customer churn is primarily driven by customers’ contract, tenure, internet service, payment method, and monthly charge. The statistical analysis also revealed some attributes have a significant relationship with churn and notable differences in customer tenure and monthly charges between customers who churned and those who did not.

The study developed and evaluated five classification models and selected the Support Vector Machine (SVM) as the best-performing algorithm. The final SVM model achieved an F1-score of 62.00%, with an accuracy of 74.59%, precision of 51.41%, recall of 78.07%, and ROC-AUC of 82.49% on the test set. Overall, the project achieved moderate performance in predicting customer churn and identified a few key drivers of churn amenable to managerial intervention. As such, the completed project can help managers make data-driven business decisions in managing and reducing customer churn rates since it provides insights into the exact probability of predicting customer churn.

This project provides empirical evidence that customer churn can be predicted using data-driven insights. As such, organizations can invest in data science to gain insights into their business data and improve customer retention. However, it is important to note that the model only identifies the likelihood of predicted outcomes and not the exact causes of the outcome.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

Based on the discovery during the exploratory analysis, statistical findings, and machine learning results, the following recommendations aim to improve customer retention:

Recommendation 1: Focus on improving early-stage retention.

It is critical to implement targeted retention initiatives and offer relevant retention services to customers at 0–12 months, as this category reported the highest churn rate of 47.44%.

Recommendation 2: Drive customers to contracts with longer terms.

Consider introducing attractive offers or benefits to customers on a month-to-month plan to move them to plans with one-year and two-year terms since the former reported a comparatively high churn rate of 42.71%.

Recommendation 3: Examine the fiber optic service offering.

The segment reported a comparatively high churn rate of 41.89%. Thus, it is critical to analyze the factors impacting customer loyalty, such as price, quality of service, and technical support.

Recommendation 4: Re-evaluate payment method options.

Since customers who pay by electronic check have the highest payment method-specific churn rate of 45.29%, it is crucial to understand their experience and consider introducing other convenient payment options.

Recommendation 5: Apply the final SVM model to prioritize customers.

The final SVM model has demonstrated a high recall rate of 78.07% and can be used to separate customers who are likely to churn from those who are not. However, the model's precision rate was 51.41%, indicating that the model may have a high false discovery rate and should be used with caution.

17.2 Future Work

The project can be extended in the following ways:

Using more customer information: The subsequent research should include such variables as customer satisfaction, complaints, service quality, technical problems, customer support, and competitors' offers to improve the quality of prediction.

More rigorous validation: The validation should not be based on an arbitrary division of data into 80% of training and 20% of testing sets. Instead, cross-validation and several repeats of train/test split may be used to obtain an unbiased estimate of model performance.

Improved model optimization: There is a need for optimizing the hyperparameters of SVM and two other classification models to find out if it is possible to improve the results.

Elimination of preprocessing leakage: During the preprocessing stage, the subsequent analysis should use the scaler obtained from the training sample to rescale the test sample. In the current study, there was a preprocessing leakage between the test and training data.

The choice of the optimal prediction threshold: Future analysis could explore different decision-function thresholds for the SVM (rather than simply using the default classification threshold). Thresholds that decreased the ratio of false positives to true positives according to the relative cost to the business of these errors would be more informative for customer-retention efforts.

Finally, **deploying the model in practice:** The ultimate goal is to implement the model in a real business process to predict at-risk customers and take measures to retain them. Therefore, the machine learning model can be embedded as a part of a business dashboard or customer management system to regularly score customers and flag at-risk ones for special attention.

18. REFERENCES

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APPENDIX A. CODE

The below appendix contains the primary code that was used for data-loading, cleaning, preprocessing, exploring data, developing the model, and evaluating the results. All the code was divided into the main steps of the project to ensure replicability.

A.1 Data Loading and Inspection

```
# Import required libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay,
    RocCurveDisplay
)
import warnings
warnings.filterwarnings('ignore')

# Set display options
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)

# Set plotting style
sns.set_theme(style='whitegrid')
print("Libraries imported successfully.")

# Load the dataset
df = pd.read_csv('Telco-Customer-Churn.csv')

# Display the first five rows
print("First five rows:")
display(df.head())

# Check dataset shape
print("\nDataset Shape:")
print(df.shape)

# Check column names
print("\nColumn Names:")
print(df.columns.tolist())

# Check data types and non-null values
print("\nDataset Information:")
df.info()

# Check missing values
print("\nMissing Values:")
```



```
print(df.isnull().sum())

# Check duplicate records
print("\nDuplicate Records:")
print(df.duplicated().sum())

# Generate descriptive statistics
print("\nDescriptive Statistics:")
display(df.describe())
```

A.2 Data Cleaning

```
# Create analysis copy
df_analysis = df.copy()

# Convert TotalCharges to numeric
df_analysis['TotalCharges'] = pd.to_numeric(
    df_analysis['TotalCharges'],
    errors='coerce'
)

# Check missing values after conversion
print("Missing Values:")
print(df_analysis.isnull().sum())

# Remove rows with missing TotalCharges, if any
df_analysis = df_analysis.dropna()

# Check duplicate records
print("Duplicate Records:", df_analysis.duplicated().sum())

# Check unique categorical values
categorical_columns = df_analysis.select_dtypes(
    include='object'
).columns

for col in categorical_columns:
    print(f"\n{col}:")
    print(df_analysis[col].unique())

# Final dataset shape
print("\nFinal Dataset Shape:", df_analysis.shape)
```

A.3 Preprocessing

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# Remove customer identifier
df_model = df_analysis.drop(columns=['customerID'])

# One-hot encode categorical variables
df_encoded = pd.get_dummies(
    df_model,
    drop_first=True
)

# Separate features and target
X = df_encoded.drop(columns=['Churn_Yes'])
y = df_encoded['Churn_Yes'].astype(int)
```

```

print("Encoded Dataset Shape:", df_encoded.shape)
print("X Shape:", X.shape)
print("y Shape:", y.shape)

# Scale numerical features
scaler = StandardScaler()

numerical_features = [
    'tenure',
    'MonthlyCharges',
    'TotalCharges'
]

X[numerical_features] = scaler.fit_transform(
    X[numerical_features]
)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training Set:", X_train.shape)
print("Testing Set:", X_test.shape)

```

Class Imbalance Handling

```

# Check target distribution
print("Training Class Distribution:")
print(y_train.value_counts())

print("\nTraining Class Proportion:")
print(y_train.value_counts(normalize=True))

# Class imbalance was handled using class_weight='balanced'
# for Logistic Regression, Decision Tree, Random Forest and SVM.

```

A.4 EDA and Visualization

1.Churn Distribution

```

plt.figure(figsize=(7, 4))

sns.countplot(
    data=df_analysis,
    x='Churn'
)

plt.title('Customer Churn Distribution')
plt.xlabel('Churn')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()

```

2.Distribution of Tenure

```

plt.figure(figsize=(7, 4))

plt.hist(

```

```
df_analysis['tenure'],
bins=20,
edgecolor='black'
)

plt.title('Distribution of Tenure')
plt.xlabel('Tenure (Months)')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```

3. Churn by Contract Type

```
plt.figure(figsize=(7, 4))

sns.countplot(
    data=df_analysis,
    x='Contract',
    hue='Churn'
)

plt.title('Customer Churn by Contract Type')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.tight_layout()
plt.show()
```

4. Tenure Distribution by Churn

```
plt.figure(figsize=(7, 4))

sns.boxplot(
    data=df_analysis,
    x='Churn',
    y='tenure'
)

plt.title('Tenure Distribution by Churn Status')
plt.xlabel('Churn')
plt.ylabel('Tenure (Months)')
plt.tight_layout()
plt.show()
```

5. Correlation Heatmap

```
plt.figure(figsize=(7, 5))

corr = df_analysis[
    [
        'tenure',
        'MonthlyCharges',
        'TotalCharges',
        'SeniorCitizen'
    ]
].corr()

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title('Correlation Heatmap of Numerical Variables')
plt.tight_layout()
plt.show()
```

A.5 Model Training and Evaluation

1. Logistic Regression

```
from sklearn.linear_model import LogisticRegression

logistic_model = LogisticRegression(
    class_weight='balanced',
    random_state=42,
    max_iter=1000
)

logistic_model.fit(X_train, y_train)

y_pred_logistic = logistic_model.predict(X_test)
```

2. Decision Tree

```
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(
    class_weight='balanced',
    random_state=42
)

dt_model.fit(X_train, y_train)

y_pred_dt = dt_model.predict(X_test)
```

3. Random Forest

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    class_weight='balanced',
    random_state=42
)

rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)
```

4. K-Nearest Neighbors

```
from sklearn.neighbors import KNeighborsClassifier

knn_model = KNeighborsClassifier()

knn_model.fit(X_train, y_train)

y_pred_knn = knn_model.predict(X_test)
```

5. Support Vector Machine

```
from sklearn.svm import SVC

svm_model = SVC(
    kernel='rbf',
    class_weight='balanced',
    random_state=42
)

svm_model.fit(X_train, y_train)

y_pred_svm = svm_model.predict(X_test)
```

6. Model Evaluation

```
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score
)

# SVM evaluation
accuracy = accuracy_score(y_test, y_pred_svm)
precision = precision_score(y_test, y_pred_svm)
recall = recall_score(y_test, y_pred_svm)
f1 = f1_score(y_test, y_pred_svm)

# SVM does not use predict_proba by default,
# so decision_function is used for ROC-AUC.
roc_auc = roc_auc_score(
    y_test,
    svm_model.decision_function(X_test)
)

print("SVM Performance")
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("ROC-AUC  :", round(roc_auc, 4))
```

7. Confusion Matrix

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm_svm = confusion_matrix(
    y_test,
    y_pred_svm
)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm_svm,
    display_labels=['No Churn', 'Churn']
)

disp.plot()

plt.title(
    'Confusion Matrix - Support Vector Machine (SVM)'
)

plt.tight_layout()
plt.show()
```

8. Training vs Test Performance

```
# Training predictions
y_train_pred_svm = svm_model.predict(X_train)

# Training performance
train_accuracy = accuracy_score(
    y_train,
    y_train_pred_svm
)

train_precision = precision_score(
    y_train,
    y_train_pred_svm
)
```

```
train_recall = recall_score(
    y_train,
    y_train_pred_svm
)

train_f1 = f1_score(
    y_train,
    y_train_pred_svm
)

# Test performance
test_accuracy = accuracy_score(
    y_test,
    y_pred_svm
)

test_precision = precision_score(
    y_test,
    y_pred_svm
)

test_recall = recall_score(
    y_test,
    y_pred_svm
)

test_f1 = f1_score(
    y_test,
    y_pred_svm
)

print("SVM Training Performance")
print(f"Accuracy : {train_accuracy:.4f}")
print(f"Precision: {train_precision:.4f}")
print(f"Recall    : {train_recall:.4f}")
print(f"F1-Score  : {train_f1:.4f}")

print("\nSVM Test Performance")
print(f"Accuracy : {test_accuracy:.4f}")
print(f"Precision: {test_precision:.4f}")
print(f"Recall    : {test_recall:.4f}")
print(f"F1-Score  : {test_f1:.4f}")
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

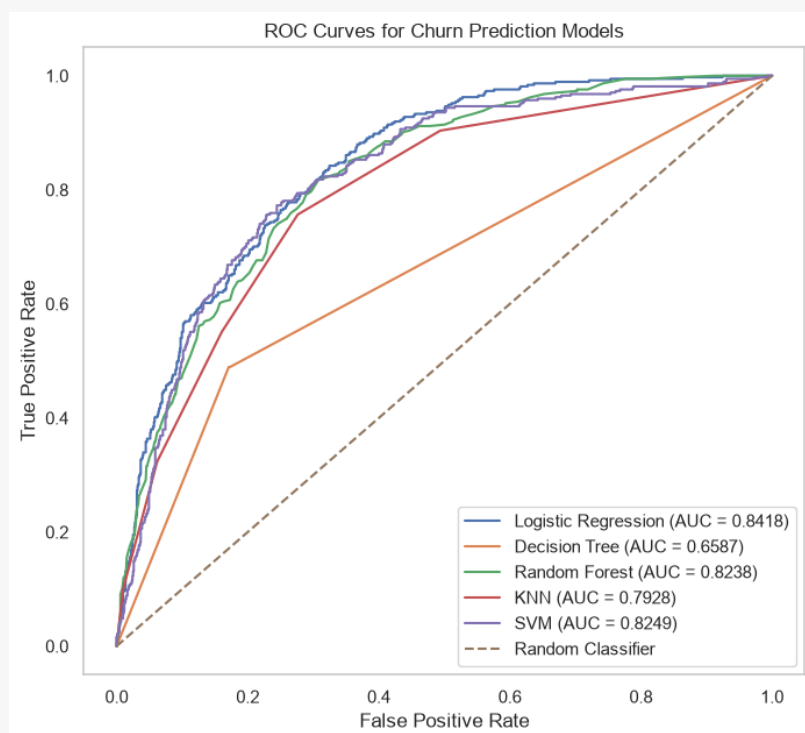


Figure 6. ROC Curves for Churn Prediction Models

Risk Factor	High-Risk Segment	Churn Rate (%)
Contract type	Month-to-month	42.71
Internet service	Fiber optic	41.89
Payment method	Electronic check	45.29
Tenure	0-12 months	47.44
Monthly charges	Medium-High	37.51

Table 7. High-Risk Customer Segments Identified from Business Analysis

Table 7 presents the customer categories with the highest percentages of churn among the studied company's business factors. The table below can be used to determine the most needed segments to be retained. Thus, the main customers to concentrate on are new, customers who pay by electronic check, month-to-month subscribers, those who choose the fiber optic option, and customers in the medium-high range for monthly charges categories.